Equilibria

CIE AS Level Chemistry (Unit 7)

- by Varaprasad

Curiosity Starters

- Why does heating **hydrated copper(II) sulfate** turn it white, but adding water turns it blue again?
- Why does ↑ pressure increase ammonia yield in the Haber process, but barely affect the Contact process?
- Why does a **catalyst** speed up equilibrium attainment but not alter equilibrium position?
- Why is **vinegar** (weak acid) much less corrosive than HCl (strong acid)?
- Why does **temperature change Kc/Kp** but concentration and pressure do not?

Chemical Equilibria

Reversible Reactions

- **Definition:** Reaction where products can reform reactants.
- **Symbol:** ⇌
- **Examples:**
  - $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$
  - $CuSO_4 \cdot 5H_2O(blue) \rightleftharpoons CuSO_4(white) + 5H_2O$

Dynamic Equilibrium

- Occurs in a **closed system**.
- **Conditions:**
  - Forward rate = Reverse rate.
  - Concentrations remain constant (not necessarily equal).

- *Analogy:* like cars entering and leaving a car park at the same rate — the number of cars inside stays the same.

Le Chatelier's Principle

If a system at equilibrium is disturbed, it shifts to **oppose the change**.

- ↑ Concentration → equilibrium shifts to use it up.
- ↑ Temperature → favours endothermic direction.
- ↑ Pressure → favours fewer moles of gas.
- Catalyst → speeds up attainment, no change in position.

Equilibrium Constants

Kc (concentration-based)

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- Only gases/aqueous appear, solids/liquids omitted.

Example:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3, \quad K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

Mole Fraction & Partial Pressure

- Mole fraction: $x_A = \frac{n_A}{n_{total}}$.
- Partial pressure: $p_A = x_A \times P_{total}$.

Kp (gas-phase reactions)

$$K_p = \frac{(p_C)^c (p_D)^d}{(p_A)^a (p_B)^b}$$

- Only gaseous species included.

Effect of Conditions on K

- **Temperature:**
  - Exothermic → ↑T, K decreases.
  - Endothermic → ↑T, K increases.
- **Concentration, Pressure, Catalyst:** do not change K.

Industrial Applications

Haber Process (ammonia, fertilisers)

$$N_2 + 3H_2 \rightleftharpoons 2NH_3, \quad \Delta H = -92\ \text{kJ/mol}$$

- 450 °C, ~200 atm, Fe catalyst.
- Compromise between yield and rate.

Contact Process (sulfuric acid)

$$2SO_2 + O_2 \rightleftharpoons 2SO_3, \quad \Delta H = -196\ \text{kJ/mol}$$

- 450 °C, 1–2 atm, V_2O_5 catalyst.
- High yield even at low pressure.

Important Exam Questions (with model answers)

Q1. Define dynamic equilibrium.
A: In a closed system, forward and reverse rates are equal, concentrations remain constant.

Q2. Why doesn't a catalyst change Kc?
A: It lowers activation energy equally in both directions.

Q3. Write Kc for $2SO_2 + O_2 \rightleftharpoons 2SO_3$.
A: $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$.

Q4. Why is 450 °C used in Haber process?
A: Lower T would give higher yield but slow rate; 450 °C is a compromise with catalyst.

Q5. Difference between strong and weak acids?
A: Strong: fully dissociate in water. Weak: partially dissociate.

Worked Example 1

1) Kc from equilibrium concentrations (simple substitution)

Question:
For the reaction

$$\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)}$$

at 500 K equilibrium concentrations are $[H_2] = 0.200\ \mathrm{mol\,dm^{-3}}$, $[I_2] = 0.200\ \mathrm{mol\,dm^{-3}}$, $[HI] = 0.600\ \mathrm{mol\,dm^{-3}}$. Calculate K_c.

Solution:

1. Write the equilibrium expression:

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

2. Substitute the equilibrium values:

$$K_c = \frac{(0.600)^2}{(0.200)(0.200)}$$

3. Compute numerator and denominator:
   - Numerator: $0.600^2 = 0.360$.
   - Denominator: $0.200 \times 0.200 = 0.040$.
4. Divide:

$$K_c = \frac{0.360}{0.040} = 9.00$$

5. Units: $(\text{mol dm}^{-3})^2/(\text{mol dm}^{-3})^2$ → dimensionless (units cancel).

Final answer: $K_c = 9.00$ (500 K).

Exam tip: Always write the formula first, that prevents exponent errors.

Try it out (Homework)

H1. For $H_2 + Cl_2 \rightleftharpoons 2HCl$ at equilibrium: $[H_2] = 0.150$, $[Cl_2] = 0.150$, $[HCl] = 0.600$. Calculate K_c.

H2. Write the K_c expression for $2NO + O_2 \rightleftharpoons 2NO_2$.

Worked Example 2

2.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

In a 2.00 dm³ vessel: $n(N_2) = 1.20,\ n(H_2) = 3.60,\ n(NH_3) = 0.$
At equilibrium, $[NH_3] = 0.600\ \text{mol dm}^{-3}$. Find [N₂], [H₂], and K_c.

Solution:

Step 1. Convert to concentrations:

$$[N_2]_0 = 1.20/2.00 = 0.600,$$
$$[H_2]_0 = 3.60/2.00 = 1.800,$$
$$[NH_3]_0 = 0.$$

Step 2. ICE Table:

Species	Initial	Change	Equilibrium
N_2	0.600	−x	0.600 − x
H_2	1.800	−3x	1.800 − 3x
NH_3	0	+2x	2x

Step 3. Use given data:

At eq, $[NH_3]$=0.600 → 2x = 0.600 → x = 0.300.

Step 4. Equilibrium concentrations:

$[N_2]=0.600 - 0.300 = 0.300,$

$[H_2]=1.800 - 0.900 = 0.900,$

$[NH_3]=0.600.$

Step 5. Kc expression and substitution:

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$K_c = \frac{(0.600)^2}{(0.300)(0.900)^3} = \frac{0.360}{0.2187} = 1.65$$

Final Answer:

$[N_2] = 0.300,\ [H_2] = 0.900,\ [NH_3] = 0.600,\ K_c = 1.65.$

Try it out (Homework)

The reaction:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

In a **1.50 dm³ vessel**:

- $n(SO_2) = 1.50\ \text{mol}$,
- $n(O_2) = 1.20\ \text{mol}$,
- $n(SO_3) = 0$.

At equilibrium, the concentration of SO_3 is $0.800\ \text{mol}\,\text{dm}^{-3}$.

Tasks:

1. Calculate the equilibrium concentrations of SO_2 and O_2.
2. Find the equilibrium constant K_c at this temperature.

Worked Example (Kp from mole fractions)

1. **Question:**

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

At equilibrium: mole fraction N_2O_4=0.40, NO_2=0.60. Total P=100 kPa. Find K_p.

Solution:

Calculate partial pressures:

Using $p_i = x_i \times P_{tot}$:

- For N_2O_4 :

$$p(N_2O_4) = 0.40 \times 100 = 40.0\ \text{kPa}$$

- For NO_2 :

$$p(NO_2) = 0.60 \times 100 = 60.0\ \text{kPa}$$

Write Kp expression and substitute the values

$$K_p = \frac{(p_{NO_2})^2}{p_{N_2O_4}}$$

$$K_p = \frac{(60.0)^2}{40.0}$$

$$K_p = \frac{3600}{40.0} = 90.0$$

Units of Kp

General rule:

$$\Delta n = (\text{gas moles products}) - (\text{gas moles reactants})$$

Here: Δn = 2 − 1 = +1.

So units = (pressure)$^{\Delta n}$ = kPa^{1}.

$$K_p = 90.0\ \text{kPa}$$

Try it out (Homework)

1. The equilibrium reaction:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

At equilibrium:

- Mole fraction of $PCl_5 = 0.25$,
- Mole fraction of $PCl_3 = 0.45$,
- Mole fraction of $Cl_2 = 0.30$.

The total pressure is $200\ \text{kPa}$.

Tasks:

1. Calculate the partial pressures of PCl_5, PCl_3, and Cl_2.
2. Write down the K_p expression for this equilibrium.
3. Calculate K_p and state its units.

Worked Example

Convert Kc → Kp (use $K_p = K_c(RT)^{\Delta n}$)

Question:

For the Haber reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

given $K_c = 0.732$ at $T = 723\ \text{K}$. Calculate K_p at this temperature. Use $R = 0.08314\ \text{L bar mol}^{-1}\ \text{K}^{-1}$. (Δn = moles gas products − reactants)

Solution

1. Δn = (2) − (1+3) = 2 − 4 = −2.
2. Use relation:

$$K_p = K_c(RT)^{\Delta n} = K_c(RT)^{-2} = \frac{K_c}{(RT)^2}$$

3. Compute RT:

$$RT = 0.08314\ \text{L bar mol}^{-1}\ \text{K}^{-1} \times 723\ \text{K} = 0.08314 \times 723 = 60.11022\ \text{L bar mol}^{-1}$$

(We are using L = dm^3; consistent with Kc in mol dm^{-3}.)

4. Square it:

$$(RT)^2 = (60.11022)^2 \approx 3613.23\ (\text{L}^2\text{bar}^2\text{mol}^{-2})$$

5. Compute K_p:

$$K_p = \frac{0.732}{3613.23} \approx 2.03 \times 10^{-4}\ \text{bar}^{-2}$$

(Units: since Δn = −2, Kp units = bar^{-2}.)

Final answer: $K_p \approx 2.03 \times 10^{-4}\ \text{bar}^{-2}$.

Exam tip: Be careful with R and units: if R in L·kPa·? use appropriate conversions; consistency is essential. If paper expects atm, convert.

Try it out (Homework)

For the equilibrium reaction:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

At **T = 673 K**, the equilibrium constant in terms of concentration is:

$$K_c = 4.20\ mol^{-1}\ dm^3$$

Tasks:

1. Calculate Δn (the change in moles of gas).
2. Write the formula linking K_c and K_p.
3. Using $R = 0.08314\ L \cdot bar \cdot mol^{-1} \cdot K^{-1}$, calculate RT.
4. Compute K_p at this temperature.
5. State the units of K_p.

Problem-Solving Strategy (Flowchart)

1. Write a balanced **equation**.
2. Decide if it's **Kc (concentration)** or **Kp (pressure)**.
3. Set up **ICE table** (Initial, Change, Equilibrium).
4. Express unknowns in terms of x.
5. Write a K expression.
6. Substitute equilibrium values.
7. Solve algebraically (may need quadratic).
8. Apply correct **units**.
9. **Check** with Le Chatelier's principle.

🧪 7.2 Brønsted–Lowry Acids & Bases

- **Acid = proton donor.**
- **Base = proton acceptor.**

Strong vs Weak:

- Strong: fully dissociate. HCl→H++Cl−HCl → H^+ + Cl^-.
- Weak: partially dissociate. CH3COOH⇌H++CH3COO−CH_3COOH ⇌ H^+ + CH_3COO^-.

pH Scale:

- pH = $-\log_{10}[H^+]$.
- 7 = neutral; <7 = acidic; >7 = alkaline.

Neutralisation:

H++OH−→H2OH^+ + OH^- → H_2O

Conjugate acid–base pair: differ by one proton.
Example: CH3COOH⇌CH3COO−+H+CH_3COOH ⇌ CH_3COO^- + H^+.

Titration Curves

- SA + SB: sharp rise at pH 7.
- WA + SB: equivalence > 7.
- SA + WB: equivalence < 7.
- WA + WB: no sharp point.

Indicators:

- Phenolphthalein → colourless in acid, pink in alkali.
- Methyl orange → red in acid, yellow in alkali.

🌟 Worked Examples (Step-by-Step)

Example 1 — Kc from concentrations

$H2+I2⇌2HIH_2 + I_2 ⇌ 2HI$

At eq: $[H_2]=0.200$, $[I_2]=0.200$, $[HI]=0.600$.

$Kc=[HI]2[H2][I2]=0.60020.200×0.200=9.00K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{0.600^2}{0.200\times0.200} = 9.00$

Example 2 — ICE + Quadratic

$2NO2⇌N2O4,Kc=6.82NO_2 ⇌ N_2O_4,\quad K_c=6.8$

Initial $[NO_2]=0.500$. ICE → solve quadratic →
$[NO2]=0.158$, $[N2O4]=0.171[NO_2]=0.158,\ [N_2O_4]=0.171$.

Example 3 — Kp from mole fractions

N2O4⇌2NO2N_2O_4 ⇌ 2NO_2

x(N₂O₄)=0.40, x(NO₂)=0.60, P=100 kPa.
pN2O4=40p_{N₂O₄}=40, pNO2=60p_{NO₂}=60.

Kp=60240=90 kPaK_p=\frac{60^2}{40}=90\ \mathrm{kPa}

Example 4 — Titration (HCl vs NaOH)

25 cm³ 0.100 M HCl titrated with 0.100 M NaOH.

- Initial pH = 1.00
- At equivalence pH = 7.00
- After 30 cm³ NaOH, excess OH^- → pH ≈ 11.96

🎯

📝 Homework Set

Section A — Easy

1. H2+Cl2⇌2HClH_2+Cl_2⇌2HCl, [H₂]=0.150, [Cl₂]=0.150, [HCl]=0.600. Find Kc.
2. Write Kc for 2NO+O2⇌2NO22NO+O_2⇌2NO_2.
3. N2O4⇌2NO2N_2O_4⇌2NO_2: p(N₂O₄)=0.40 atm, p(NO₂)=0.60 atm. Find Kp.
4. 0.20 mol CO + 0.20 mol H_2O ⇌ 0.10 mol CO_2 + 0.10 mol H_2. Find Kc.

5. Why does ↑T decrease K for exothermic reactions?

Section B — Medium

6. 1.00 mol PCl_5 in 2 dm^3 → 0.40 mol decomposed. Find Kc.
7. In 1 dm^3, $[NH_3]=0.200$, $[N_2]=0.300$, $[H_2]=0.900$. Find Kc.
8. At 150 kPa, $y(SO_2)=0.30$, $y(O_2)=0.20$, $y(SO_3)=0.50$. Find Kp.
9. Why are catalysts used in industry if they don't change K?
10. 2NO2⇌N2O4, Kc=6.82$NO_2 \rightleftharpoons N_2O_4$,\ K_c=6.8. If $[NO_2]_0=0.80$, find equilibrium [ ].

Section C — Hard

11. CO+H2O⇌CO2+H2, Kc=4.0$CO+H_2O \rightleftharpoons CO_2+H_2$,\ K_c=4.0. Start $[CO]=[H_2O]=0.25$. Find eq [ ].
12. N2+O2⇌2NO$N_2+O_2 \rightleftharpoons 2NO$. Given mole fractions, P=1 atm, $Kp=2.0\times10^{-5}$. Check equilibrium.
13. Explain quantitatively why ↑P improves NH_3 yield in Haber.
14. Titrate 20 cm^3 0.150 M CH_3COOH with 0.100 M NaOH. Find NaOH vol at equivalence & pH at half-eq.
15. 0.80 mol SO_2 + 0.50 mol O_2 in 2 dm^3 ⇌ 0.36 mol SO_3 at eq. (a) Find [ ]. (b) Calculate Kc. (c) If P=200 kPa, find Kp.

🔑 Homework Answer Highlights

- **Q1:** Kc=16.0
- **Q3:** Kp=0.90 atm
- **Q6:** Kc=0.133 mol dm^{-3}
- **Q7:** Kc≈0.183
- **Q10:** $[NO_2]=0.276$, $[N_2O_4]=0.262$
- **Q11:** $[CO]=[H_2O]=0.0833$, $[CO_2]=[H_2]=0.1667$
- **Q12:** Qp ≫ Kp → too much NO, shift left
- **Q14:** V=30.0 cm^3, pH=pKa at half-eq
- **Q15:** $[SO_2]=0.220$, $[O_2]=0.160$, $[SO_3]=0.180$, Kc≈4.18, Kp≈0.0118 kPa^{-1}

⚡ Rapid-Fire Glossary

- **Dynamic equilibrium:** forward = reverse rates, closed system.
- **Le Chatelier's principle:** system shifts to oppose change.
- **Kc:** equilibrium constant (concentrations).
- **Kp:** equilibrium constant (pressures).
- **Mole fraction:** part/whole of moles.
- **Partial pressure:** mole fraction × total P.
- **Exothermic:** heat out (ΔH < 0).

- **Endothermic:** heat in ($\Delta H > 0$).
- **Strong acid/base:** fully dissociates.
- **Weak acid/base:** partially dissociates.
- **Neutralisation:** $H^+ + OH^- \rightarrow H_2O$.
- **Indicator:** shows colour at equivalence point.

✅ **All in one place now**:

- Hooks
- Clear definitions
- Theory & formulas
- Flowchart for problem-solving
- Worked examples
- Exam-style Q&As
- Homework + solutions
- Glossary

Would you like me to also **turn this into a 2-column printable revision handout** (left = key facts/formulas, right = examples & tips)? That's a format many top schools use for exam prep.

Practice questions

1. The reaction

$$\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)}$$

At equilibrium at 500 K the concentrations are $[H_2] = 0.200\ \mathrm{mol\ dm^{-3}}$, $[I_2] = 0.200\ \mathrm{mol\ dm^{-3}}$, $[HI] = 0.600\ \mathrm{mol\ dm^{-3}}$.

Calculate the equilibrium constant Kc.

2. For the reaction

$$\mathrm{2\,SO_3(g) \rightleftharpoons 2\,SO_2(g) + O_2(g)}$$

Write the expression for K_c.

3. For the equilibrium

$$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$$

At equilibrium: $[H_2] = 0.200$, $[Cl_2] = 0.200$, $[HCl] = 0.800$ mol dm^{-3}.

Calculate **Kc**.

4. For the equilibrium

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

at equilibrium, partial pressures are: $p_{N_2O_4} = 0.50 \text{ atm}, \; p_{NO_2} = 0.80 \text{ atm}.$

Calculate **Kp**.

5. 1.00 mol of HI is placed in a 2.00 dm^3 vessel:

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

At equilibrium, 0.25 mol HI has decomposed.

Calculate **Kc**.

6. In a 1.00 dm^3 vessel:

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

Initial concentrations: [CO] = [H_2O] = 0.400 mol dm^{-3}. At equilibrium [H_2] = 0.150 mol dm^{-3}.

Find **Kc**.

7. For the Haber process:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

At equilibrium in a 1.00 dm^3 container, there are 0.20 mol N_2, 0.40 mol H_2, and 0.60 mol NH_3. Calculate **Kc**.

8. One mole of PCl_5 is placed in a 2.00 dm^3 vessel and allowed to come to equilibrium:

$$\mathrm{PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)}$$

At equilibrium 0.60 mol PCl_5 remains. **Calculate** K_c.

9. At 1000 K the equilibrium mixture in a 1.00 dm³ vessel for

$$\mathrm{CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)}$$

contains $n(CO) = 0.20,\ n(H_2O) = 0.30,\ n(CO_2) = 0.40,\ n(H_2) = 0.10$ mol. **Calculate K_c and state qualitatively what happens if more CO is added.**

10. At a certain temperature the equilibrium mixture for

$$\mathrm{N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)}$$

has mole fractions $y_{N_2} = 0.25,\ y_{H_2} = 0.50,\ y_{NH_3} = 0.25$ at total pressure $P_{tot} = 200\ \mathrm{kPa}$. **Calculate K_p (include units).**